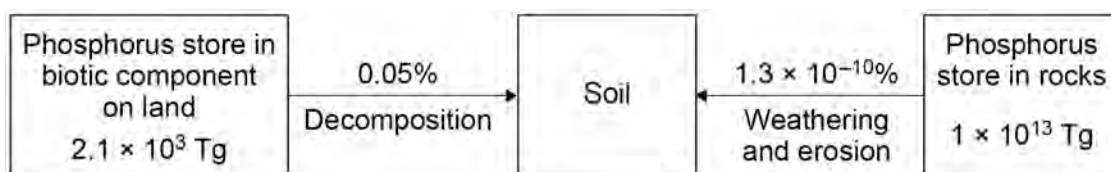


0 3

Figure 1 shows part of a phosphorus cycle, including the mass of phosphorus in two stores.

Figure 1



One teragram (Tg) is 1×10^{12} grams.

The supply of phosphorus to the soil is shown as a percentage of the total phosphorus mass of each store.

0 3 . 1

Calculate in teragrams the difference in the supply of phosphorus to the soil from the two stores shown in **Figure 1**.

Show your working.

[2 marks]

Answer _____ Tg

Question 3 continues on the next page

Water shortage can inhibit crop production. Scientists investigated the effect of the mycorrhizal species *Glomus intraradices* on the growth of tomato plants under conditions of water shortage and **no** water shortage.

- The scientists planted tomato seeds into a large number of pots containing sterilised soil.
- They added a culture of *G. intraradices* to 50% of the pots and 50% were left untreated.
- After the seeds had developed into seedlings (young plants), the seedlings from the untreated and treated pots were planted into four separate large greenhouses (glasshouses).
- A very limited supply of water (water shortage) was provided to two of the greenhouses. A sufficient supply of water (**no** water shortage) was provided to the other two greenhouses.
- After 60 days, the scientists determined the mean mass of tomatoes (kg m^{-2}) from each greenhouse.

Greenhouse **P** – untreated seedlings with water shortage

Greenhouse **Q** – treated seedlings with water shortage

Greenhouse **R** – untreated seedlings with **no** water shortage

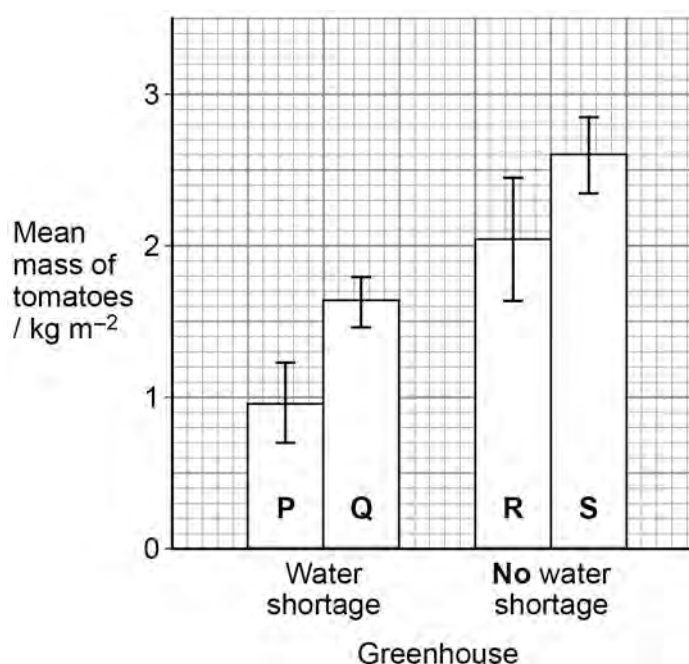
Greenhouse **S** – treated seedlings with **no** water shortage

Treated seedlings are plants grown with *G. intraradices*.

Figure 2 shows the scientists' results.

The error bars represent ± 2 standard deviations from the mean, which includes over 95% of the data.

Figure 2



[5 marks]

[illegible]

0 3 . 3

The tomato seeds were planted into sterilised soil.

Suggest **two** reasons why the soil was sterilised.

[2 marks]

1 _____

2 _____

0 3 . 4

The soil used in this investigation had the recommended concentration of fertiliser.

Suggest **two** reasons why the soil contained the recommended concentration of fertiliser.

[2 marks]

1 _____

2 _____

1

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

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